

Question 2 - Instruction Set 2 – x295+ - SOLUTION

B. Compiling and assembling a C program using our x295+ instruction set

Table 1

C program (The C code below cannot be modified – it must stay as it is stated below)	x295+ assembly program	x295+ machine code program
$z = (x + y) * (x - y);$	LOAD x, r0 LOAD y, r1 ADD r0, r1, r2 SUB r0, r1, r3 MUL r2, r3, r4 STORE r4, z	1010 000 XXX XXXXXX 0000 <Src1 12 bits> 1010 001 XXX XXXXXX 0000 <Src2 12 bits> 0001 010 000 001 XXX 0010 011 000 001 XXX 0011 100 010 011 XXX 1011 XXX 100 XXXXXX 0000 <Dest 12 bits>

C. Evaluating our x295+ instruction set using Memory Traffic criteria

Table 2

x295+ program (1 assembly instruction/ machine code instruction per row)	Fetch (number of memory accesses of word size) + Explain how you obtain your count	Decode/Execute (number of memory accesses of word size) + Explain how you obtain your count
Assembly instruction: LOAD x, r0 Machine code: 1010 000 XXX XXXXXX 0000 <Src1 12 bits>	Count: 2 Explanation: fetching 1 instruction that is 2-word wide	Count: 1 Explanation: executing a LOAD requires 1 memory access -> reading value x (16 bits -> 1 word) from memory
Assembly instruction: LOAD y, r1 Machine code: 1010 001 XXX XXXXXX 0000 <Src2 12 bits>	Count: 2 Explanation: fetching 1 instruction that is 2-word wide	Count: 1 Explanation: executing a LOAD requires 1 memory access -> reading value y (16 bits -> 1 word) from memory
Assembly instruction: ADD r0, r1, r2 Machine code: 0001 010 000 001 XXX	Count: 1 Explanation: fetching 1 instruction that is 1-word wide	Count: 0 Explanation: executing an ADD does not require memory access since it is manipulating values held in registers
Assembly instruction: SUB r0, r1, r3 Machine code: 0010 011 000 001 XXX	Count: 1 Explanation: fetching 1 instruction that is 1-word wide	Count: 0 Explanation: executing an SUB does not require memory access since it is

		manipulating values held in registers
Assembly instruction: MUL r2, r3, r4 Machine code: 0011 100 010 011 XXX	Count: 1 Explanation: fetching 1 instruction that is 1-word wide	Count: 0 Explanation: executing an MUL does not require memory access since it is manipulating values held in registers
Assembly instruction: STORE r4, z Machine code: 1011 XXX 100 XXXXXX 0000 <Dest 12 bits>	Count: 2 Explanation: fetching 1 instruction that is 2-word wide	Count: 1 Explanation: executing a STORE requires 1 memory access -> writing value z (16 bits -> 1 word) to memory
Grand Total: 12 memory accesses	Total: 9 memory accesses	Total: 3 memory accesses

Once completed, submit it on Crowdmark as your answer to Question 2.